In [4]:
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#EXPLORATORY DATA ANALYSIS (EDA)
# PROJECT: Intelligent Network Intrusion Detection
# Dataset File Name: pbl_dataset.csv
# Main Features: Packet_Size, Connection_Duration, Error_Rate, Label

#IMPORT REQUIRED LIBRARIES

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# To display all columns properly
pd.set_option("display.max_columns", None)

#LOAD THE DATASET
df = pd.read_csv("pbl_dataset.csv")
print("DATASET LOADED SUCCESSFULLY")

#DATA UNDERSTANDING
# First 5 rows
print("\nFIRST 5 ROWS")
print(df.head())

# Last 5 rows
print("\nLAST 5 ROWS")
print(df.tail())

# Shape of dataset
print("\nSHAPE OF DATASET")
print("Total Rows:", df.shape[0])
print("Total Columns:", df.shape[1])

# Column names
print("\nCOLUMN NAMES")
print(df.columns)

# Data types
print("\nDATA TYPES")
print(df.dtypes)

# Complete dataset info
print("\nDATASET INFO")
print(df.info())

#CHECK FOR MISSING VALUES
print("\nMISSING VALUES")
print(df.isnull().sum())

# Visual representation of missing values
plt.figure(figsize=(10,5))
sns.heatmap(df.isnull(), cbar=False, cmap="viridis")
plt.title("Missing Values Heatmap")
plt.show()

#CHECK FOR DUPLICATE VALUES
print("\nDUPLICATE VALUES ")
duplicate_count = df.duplicated().sum()
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print("Total Duplicate Rows:", duplicate_count)


#BOXPLOTS FOR OUTLIER DETECTION
# Packet Size
if 'Packet_Size' in df.columns:
    plt.figure(figsize=(8,4))
    sns.boxplot(x=df['Packet_Size'])
    plt.title("Boxplot of Packet Size")
    plt.show()

# Connection Duration
if 'Connection_Duration' in df.columns:
    plt.figure(figsize=(8,4))
    sns.boxplot(x=df['Connection_Duration'])
    plt.title("Boxplot of Connection Duration")
    plt.show()

# Error Rate
if 'Error_Rate' in df.columns:
    plt.figure(figsize=(8,4))
    sns.boxplot(x=df['Error_Rate'])
    plt.title("Boxplot of Error Rate")
    plt.show()

#LABEL ANALYSIS (NORMAL VS INTRUSION)
    plt.figure(figsize=(6,4))
    sns.countplot(x='Label', data=df)
    plt.title("Normal Traffic vs Intrusion Traffic")
    plt.xlabel("Label (0 = Normal, 1 = Intrusion)")
    plt.ylabel("Count")
    plt.show()
    print("\nLABEL DISTRIBUTION")
    print(df['Label'].value_counts())

#CATEGORICAL ANALYSIS
categorical_columns = df.select_dtypes(include='object').columns
for column in categorical_columns:
    plt.figure(figsize=(8,4))
    sns.countplot(x=df[column])
    plt.title("Count Plot of " + column)
    plt.xticks(rotation=45)
    plt.show()

#GROUPED ANALYSIS
# Example: Average Error Rate by Label
if 'Error_Rate' in df.columns and 'Label' in df.columns:
    grouped_data = df.groupby("Label")["Error_Rate"].mean()
    print("\nAVERAGE ERROR RATE BY LABEL ")
    print(grouped_data)
    grouped_data.plot(kind='bar', figsize=(8,4))
    plt.title("Average Error Rate by Label")
    plt.ylabel("Average Error Rate")
    plt.show()

#FINAL OBSERVATIONS
print("\nINAL OBSERVATIONS")
print("1. Missing values and duplicates were identified.")
print("2. Distribution of important network features was analyzed.")
print("3. Outliers were detected using boxplots.")
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print("4. Normal and malicious traffic were compared.")
print("5. Relationships between network parameters were visualized.")

#SAVE EDA PROCESSED FILE
df.to_csv("eda_dataset.csv", index=False)

print("\n========== EDA COMPLETED SUCCESSFULLY ==========")
print("EDA processed dataset saved as: eda_dataset.csv")
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DATASET LOADED SUCCESSFULLY

FIRST 5 ROWS
   Packet_Size  Connection_Duration  Error_Rate Protocol  Source_Port  \
0          402                    5        0.20      TCP        10459
1          735                    1        0.84     ICMP        38790
2          570                    5        0.82      UDP         1186
3          406                    5        0.64      TCP        35766
4          371                    2        0.25      TCP        20699

   Destination_Port  Bytes_Sent  Bytes_Received  Login_Attempts  \
0                20       18090            4422               1
1                53        7350           10290               9
2                20        8550            3990               8
3                25       17052            2030               1
4               143        3710            4823               1

   Failed_Logins Encryption_Type  Label
0              1            WPA2      0
1              3             AES      1
2              3            WPA2      1
3              1            WPA2      0
4              1            WPA2      0

LAST 5 ROWS
      Packet_Size  Connection_Duration  Error_Rate Protocol  Source_Port  \
1995          704                    3        0.94      UDP        20605
1996          545                    3        0.97     ICMP        49716
1997          890                    2        0.61      TCP        60746
1998          770                    1        0.56      TCP        27524
1999          722                    1        0.42      TCP        33050

      Destination_Port  Bytes_Sent  Bytes_Received  Login_Attempts  \
1995                25       32384           14784              12
1996                20        5450           10900               9
1997               143       20470           29370               8
1998                21        5390           26950              14
1999               143       15884           14440              13

      Failed_Logins Encryption_Type  Label
1995              3         SSL/TLS      1
1996              3         SSL/TLS      1
1997              6             NaN      1
1998              4            WPA2      1
1999              6         SSL/TLS      1

SHAPE OF DATASET
Total Rows: 2000
Total Columns: 12

COLUMN NAMES
Index(['Packet_Size', 'Connection_Duration', 'Error_Rate', 'Protocol',
       'Source_Port', 'Destination_Port', 'Bytes_Sent', 'Bytes_Received',
       'Login_Attempts', 'Failed_Logins', 'Encryption_Type', 'Label'],
      dtype='object')

DATA TYPES
Packet_Size              int64
Connection_Duration      int64
Error_Rate             float64
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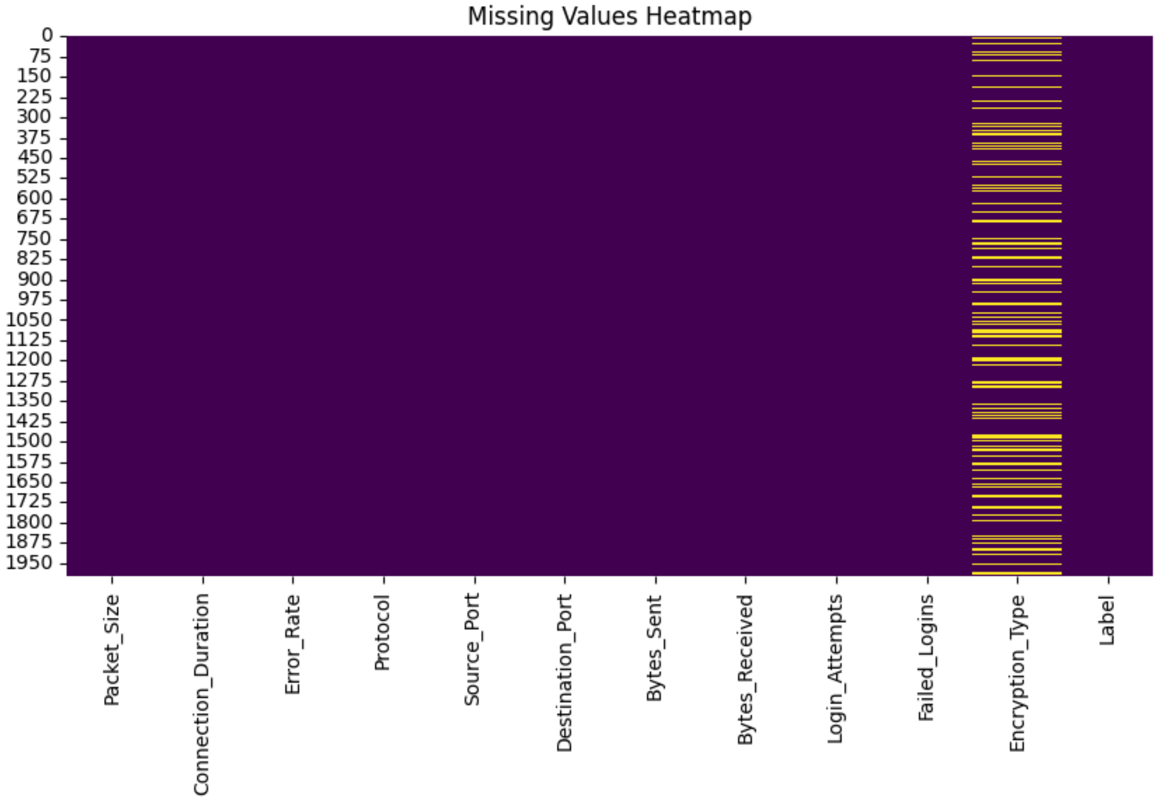
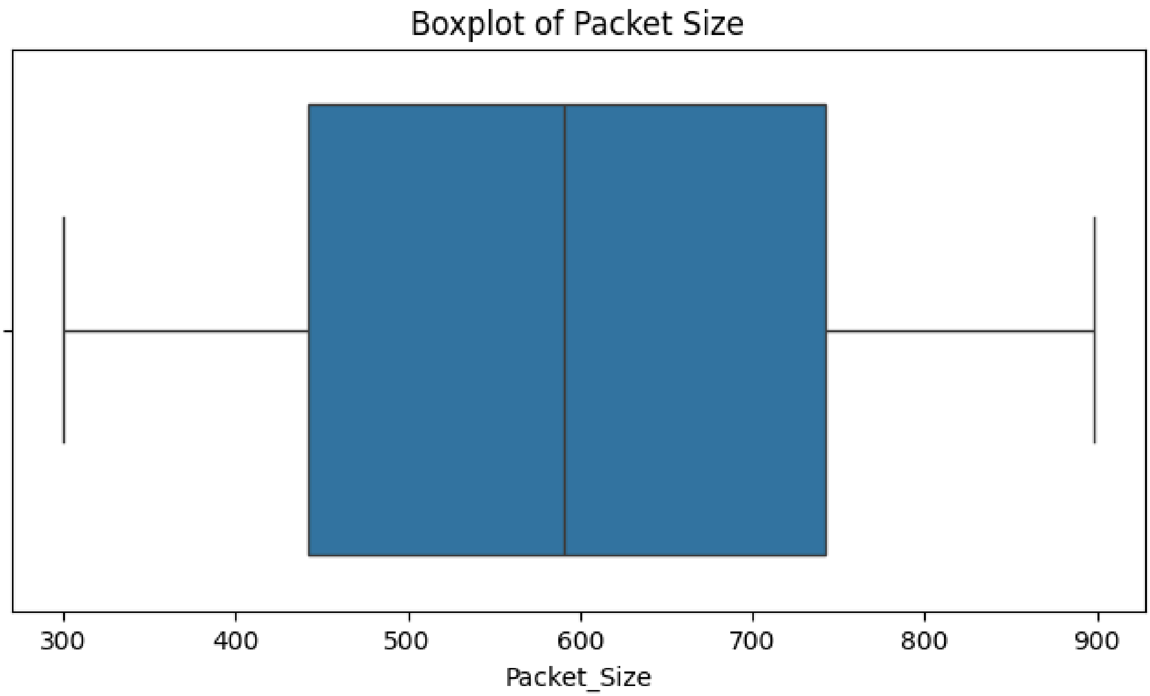
Protocol                object
Source_Port              int64
Destination_Port         int64
Bytes_Sent               int64
Bytes_Received           int64
Login_Attempts           int64
Failed_Logins            int64
Encryption_Type         object
Label                    int64
dtype: object

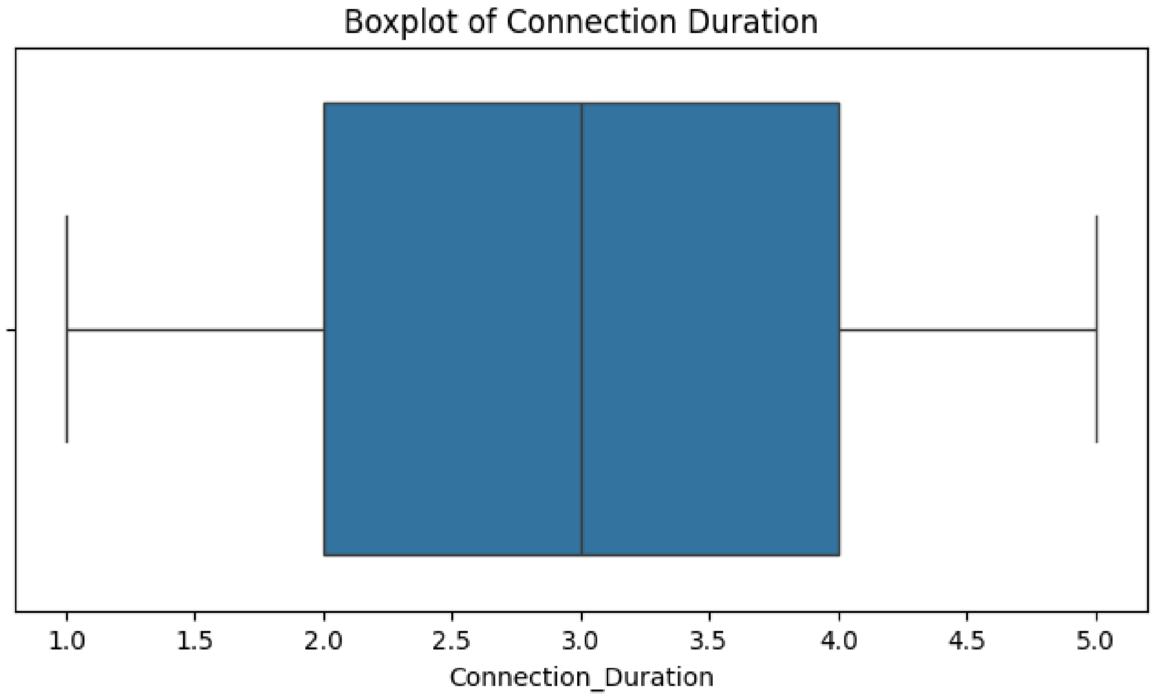
DATASET INFO
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 12 columns):
 #   Column               Non-Null Count  Dtype  
---  ------               --------------  -----  
 0   Packet_Size          2000 non-null   int64  
 1   Connection_Duration  2000 non-null   int64  
 2   Error_Rate           2000 non-null   float64
 3   Protocol             2000 non-null   object 
 4   Source_Port          2000 non-null   int64  
 5   Destination_Port     2000 non-null   int64  
 6   Bytes_Sent           2000 non-null   int64  
 7   Bytes_Received       2000 non-null   int64  
 8   Login_Attempts       2000 non-null   int64  
 9   Failed_Logins        2000 non-null   int64  
 10  Encryption_Type      1491 non-null   object 
 11  Label                2000 non-null   int64  
dtypes: float64(1), int64(9), object(2)
memory usage: 187.6+ KB
None

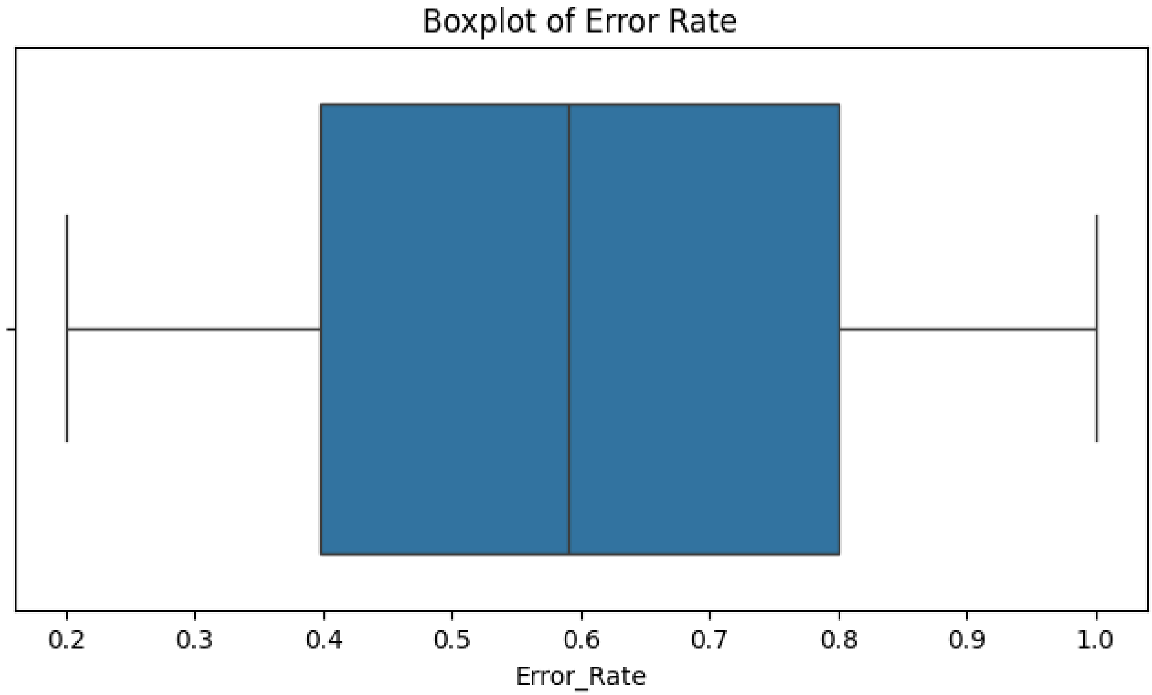
MISSING VALUES
Packet_Size              0
Connection_Duration      0
Error_Rate               0
Protocol                 0
Source_Port              0
Destination_Port         0
Bytes_Sent               0
Bytes_Received           0
Login_Attempts           0
Failed_Logins            0
Encryption_Type        509
Label                    0
dtype: int64
```

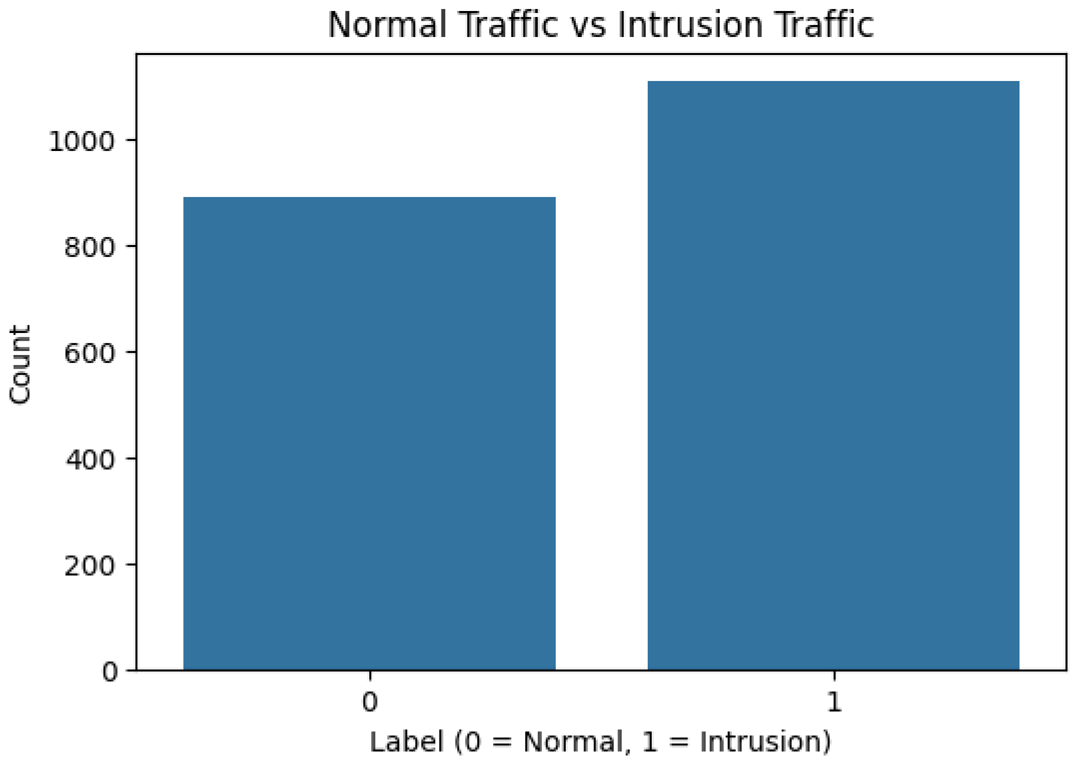



```
DUPLICATE VALUES
Total Duplicate Rows: 0
```




Boxplot of Connection Duration
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Connection_Duration


Boxplot of Error Rate
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Error_Rate

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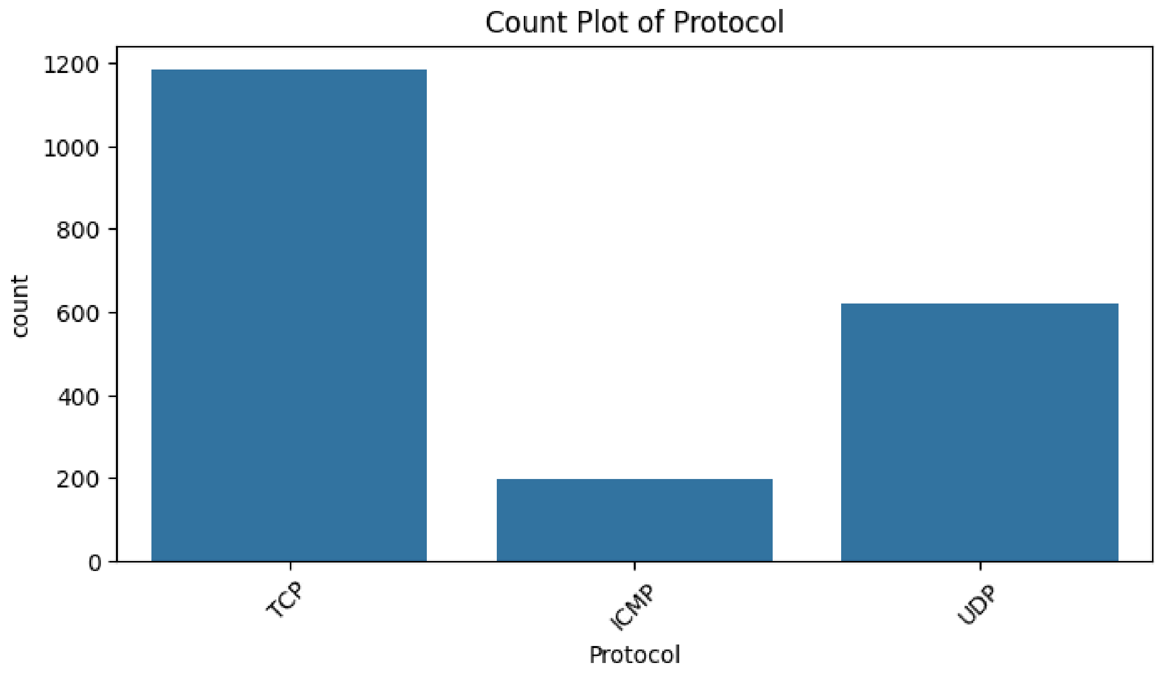
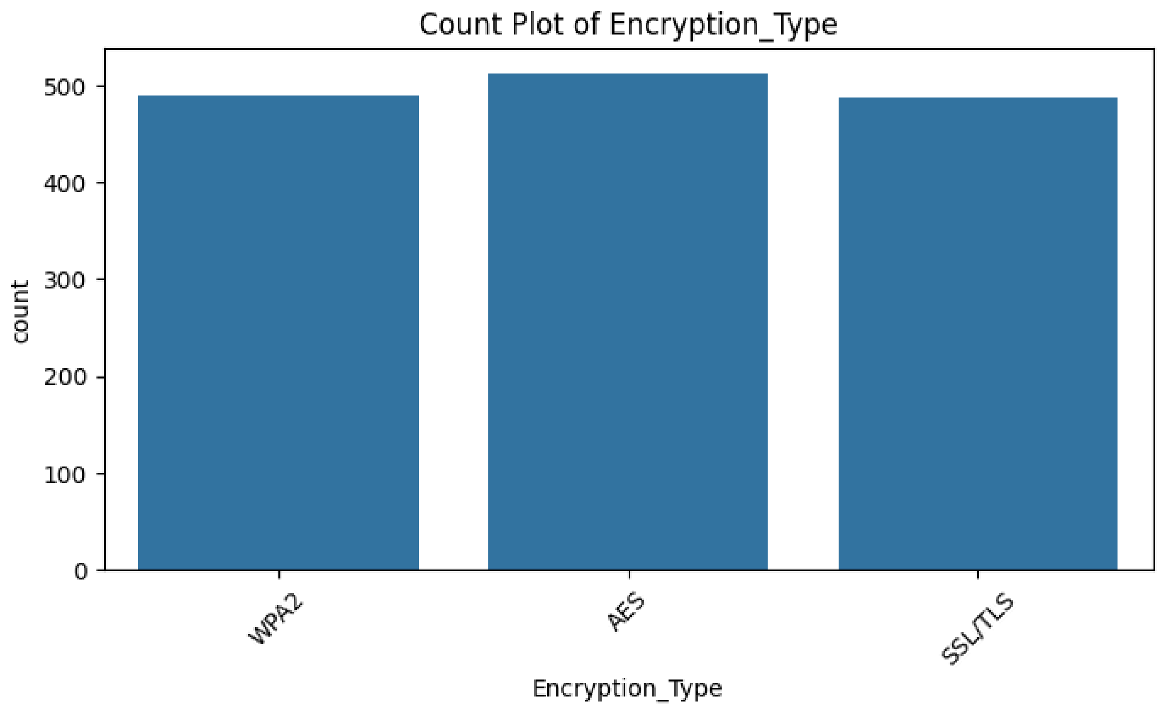
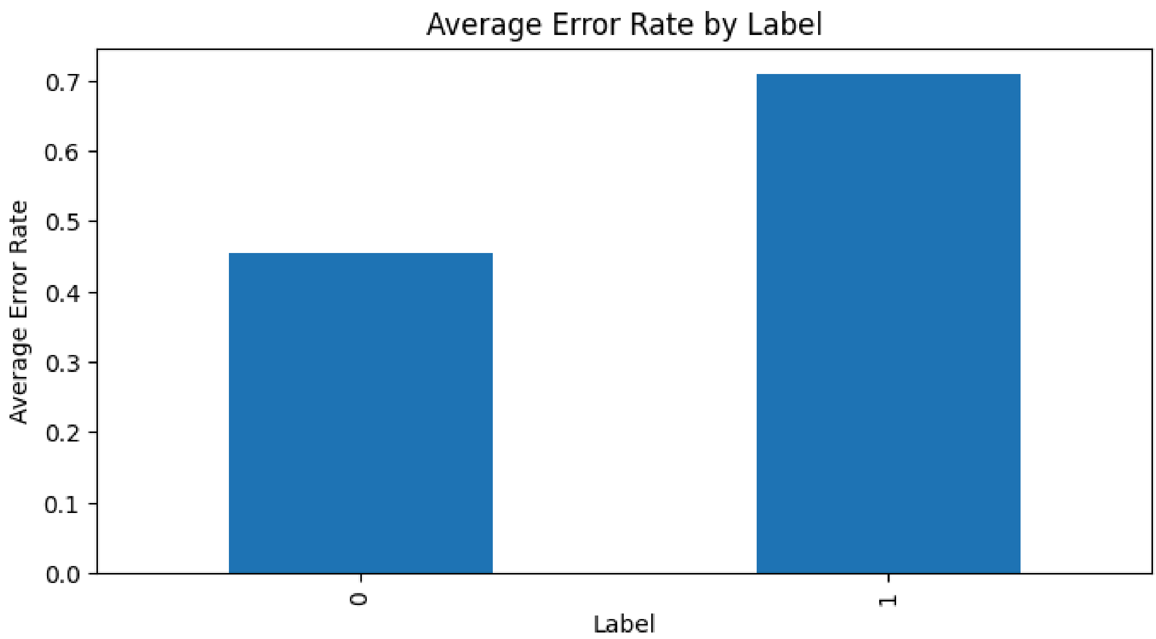
LABEL DISTRIBUTION
Label
1    1109
0     891
Name: count, dtype: int64
```






```
AVERAGE ERROR RATE BY LABEL
Label
0    0.454366
1    0.710009
Name: Error_Rate, dtype: float64
```




```
INAL OBSERVATIONS
1. Missing values and duplicates were identified.
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4. Normal and malicious traffic were compared.
5. Relationships between network parameters were visualized.

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In [ ]: